MIL - STD- 1870(AT)

13 May 1983

Superseding
(see 6.1)

MILITARY STANDARD

GRAIN SIZE, CAST TURBINE BLADES
ACCEPTANCE CRITERIA FOR

FSC 2835

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MIL-STD-1870(AT)
Department of Defense
Washington, D.C. 20301

Grain Size, Cast Turbine Blades, Acceptance Criteria for

1. This Military Standard is approved for use by U.S. Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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FOREWORD

The intent of this standard is to define grain size acceptance criteria for cast turbine blades.

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1. SCOPE

1.1 Scope. The procedures covered by this standard are to be used to insure cast turbine blades meet prescribed grain size acceptance criteria.

1.2 Classification. Castings shall be classified in accordance with grain size requirements of table I. Castings not listed in table I by part number shall be suitably classified on applicable engineering drawing.

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2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this standard to the extent specified herein.

SPECIFICATIONS
MILITARY

M1PD-N-62441	- Casting, Investment, Nickel Base (713C)
M1PD-N-62443	- Nickel Base Alloy, Castings, Investment
M1PD-N-62444	- Nickel Base Alloy, Castings, Investment (713LC)
M1PD-N-62445	- Nickel Base Alloy, Castings, Investment (0021)
M1PD-N-62446	- Nickel Base Alloy, Castings, Investment (C101)
M1PD-N-62447	- Nickel Base Alloy, Castings, Investment (C103)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the Army Tank-Automotive Command, ATTN: DRSTA-GSS Warren, MI 48090, or as directed by the contracting officer.)

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3. DEFINITIONS

3.1 Definitions. For the purposes of this standard, the following definitions shall apply.

3.1.1 Grain size. The measurements made to determine grain size are made on surface of casting. Dimensional limitations specified herein are defining grain structure as a surface phenomena. Internal grain size is excluded from requirements specified in 4.1. However, it is expected that a grain occurring at the juxtaposition of two or more planes will be measured separately in each plane where it appears.

3.1.2 Airfoil shape. An airfoil shape is a casting or portion of a casting which extracts work or changes direction of the hot gas stream in the turbine section of a gas turbine engine.

3.1.3 Other shapes. Other shapes are any casting or portion of a casting which cannot be defined as an airfoil shape (e.g.: root area or tip shroud of a cast blade).

3.1.4 Casting lot. Unless otherwise specified in applicable process specifications, a casting lot is a group of castings with the same part number and revision letter poured from one master heat of metal using the same casting process parameters and the same type of preheat and casting furnace.

3.1.5 Columnar grains. A columnar grain shall be any grain which has its maximum dimension at least four times the length of its minimum dimension. Any columnar grain 0.125 inch (3.18 millimeter (mm)) maximum dimension or less shall not be considered ratable.

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4. GENERAL REQUIREMENTS

4.1 Acceptance criteria. Unless otherwise specified in table I, grain size shall not exceed the following requirements:

4.1.1 Airfoil shapes. The grain size shall be generally equiaxed and uniform throughout the airfoil (see 3.1.2) and shall not exceed maximum dimension for the applicable material:

a. M1PD-N-62441 and M1PD-N-62444 material. 0.25 inch (6.35 mm) Maximum dimension.
b. M1PD-N-62443, M1PD-N-62445, M1PD-N-62446 and M1PD-N-62447 material. 0.125 inch (3.18 mm) maximum dimension.

4.1.2 Other shapes. The grain size in other cast shapes (see 3.1.3) shall be generally equiaxed and uniform throughout the section and shall not exceed the following:

a. M1PD-N-62441 and M1PD-N-62444 material. 0.250 inch (6.35 mm) maximum dimension.
b. M1PD-N-62443, M1PD-N-62445, M1PD-N-62446 and M1PD-N-62447 material:

Casting thickness Inch (mm)	Grain size Inch (mm)
To 0.125 (3.18) incl.	0.125 (3.18)
Over 0.125 (3.18 to 0.500 (12.70) incl.	0.187 (4.75)
Over 0.500 (12.70)	0.250 (6.35)

4.1.3 Columnar grains. Unless otherwise specified in table I, columnar grains (see 3.1.5) shall not be acceptable.

4.1.4 Transition zones. Unless otherwise specified on the engineering drawing or table I, sharp areas of transition in grain size shall not be cause for rejection.

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5. DETAIL INSPECTION REQUIREMENTS

5.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own facilities or any other facility acceptable to the procuring activity. The procuring activity reserves the right to perform or witness any of the necessary tests to assure that the product conforms to the requirements of this standard.

5.2 Reports. Unless otherwise specified in purchase order, supplier shall furnish (with each shipment) 3 copies of a certificate stating the following:

a. Purchase order number.

b. Part number and revision letter.

c. Quantity.

d. Conformance to MIPD-STD-1870, latest revision.

5.3 Frequency of inspection. A minimum of the first 500 blade castings from every casting lot (see 3.1.4) shall be grain size inspected. If first 500 blade castings are acceptable, the balance of the casting lot may be inspected with a sample plan that has been approved by the acquisitioning activity.

5.4 Test method. Each casting shall be 100 percent grain size etched to verify conformance to 4.1. Etchant used shall be approved by procuring activity prior to use. 1/

1/ After completion of grain size inspection, etched castings may be cleaned by light abrasive blasting.

6. NOTES

6.1 Supersession. This military standard supersedes AVCO Lycoming Division, internal specification P7075B, dated 5 October 1979.

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TABLE I. Grain size requirements.

Classification	Part number	Requirements
MIL-STD-1870-P1	0-121-041-01 0-121-041-02 0-129-001-07 0-141-002-01 0-141-004-01 0-141-006-01 0-141-008-01 0-141-043-01 0-141-043-03 1-100-064-06 1-100-118-08 1-100-127-02 1-100-256-02 1-100-372-08 1-100-428-15 1-140-065-28 1-141-211-13 1-140-211-14 1-140-273-02 1-190-007-04 2-120-001-76 2-121-041-21 2-121-041-22 2-121-041-27 2-121-041-28 2-121-043-09 2-121-043-10 2-121-043-21 2-121-043-22 2-121-139-01 2-121-139-02 2-140-001-41 2-140-001-57 2-140-002-40 2-141-034-01 2-141-034-02 2-141-036-47	a. Columnar grains shall not exceed the following in the areas designated in figure 1: Areas 1 and 2 - Columnar grains up to 0.50 inch (12.7 mm) maximum dimension not exceeding an angle of 45° to the stacking line shall be permitted. Not more than 1/5 of this area shall contain columnar grains. Area 3 - No ratable columnar grains shall be permitted. Area 4 - Columnar grains up to 0.37 inch (9.5 mm) maximum dimension shall be permitted. Area 5 - No ratable columnar grains shall be permitted.

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TABLE I. Grain size requirements. - Continued

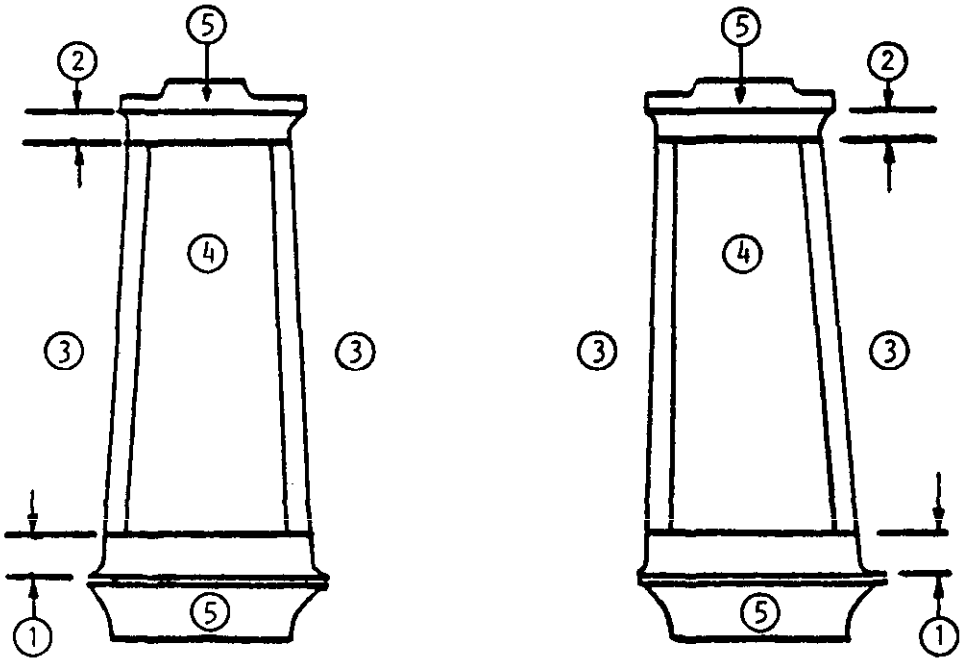
Classification	Part number	Requirements
MIL-STD-1870-P2	0-141-002-03 0-141-008-02 1-190-007-08 2-120-001-89 2-141-034-29 2-141-034-35	a. Airfoil grains shall be in accordance with 4.1.1 except for area 3 which shall have maximum equiaxed grains as follows: Maximum dimensions for MIPD-N-62441 and MIPD-N-62444 alloys shall be 0.125 inch (3.175 mm) maximum dimension. All other alloys shall be 0.075 inch (1.90 mm) maximum dimension. b. Columnar grains shall not exceed the following in the areas designated in figure 1: Areas 1 and 2 - Columnar grains up to 0.31 inch (7.8 mm) maximum dimension not exceeding an angle of 45° to the stacking line shall be permitted. Area 3 - No ratable columnar grains shall be permitted. Area 4 - Columnar grains up to 0.25 inch (6.3 mm) maximum dimension shall be permitted. Not more than 1/5 of this area shall contain columnar grains. Area 5 - No ratable columnar grains shall be permitted.

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TABLE I. Grain size requirements. - Continued

Classification	Part number	Requirements
MIL-STD-1870-P3	2-121-142-01 2-121-142-02 2-121-142-07 2-121-142-08	a. Columnar grains up to 0.375 inch (9.53 mm) shall be acceptable. b. Columnar grains shall not intersect the airfoil trailing edge from a point 0.375 inch (9.53 mm) above the shroud platform.

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Casting Areas:

1 15 percent of blade length at the trailing edge.

2 20 percent of blade length at the trailing edge.

3 10 percent of mean chordal width.

4 Airfoil surface as shown.

5 See 4.1.2

FIGURE 1. Casting areas for grain size allowance.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-STD-1870(AT)	Grain Size, Cast Turbine Blades, Acceptance Criteria for

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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